



East West University

Lab Report

Semester: Summer-2025

Course Title: Electronic Circuits **Course Code:** CSE251

Section: 04

Expt No: 07

Expt Name: Biasing of a Common-Source Voltage Amplifier

Group No: 06

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Date of Performance: August 18, 2025

Date of Submission: August 25, 2025

Title: Biasing of a Common-Source Voltage Amplifier

Objectives

The objective is to identify an appropriate DC operating point for a NMOS transistor.

Theory (Summary)

This experiment explains the operation of a common-source amplifier using an NMOS transistor. It details the biasing process, which involves fixing the gate voltage with a voltage divider and using a source resistor (R_S) to stabilize the bias current. The source resistor provides negative feedback, helping to maintain consistent performance despite temperature variations and transistor characteristics. The text also discusses selecting resistors to ensure the drain voltage (V_{DS}) is about half of (V_{DD}), allowing for a large and symmetrical output swing. Additionally, it highlights the importance of transconductance (g_m), which measures the rate of change of the output current concerning the input voltage (V_{GS}). Equations are provided to calculate the drain current (i_D) and transconductance (g_m).

Equipment and Component

1. Digital trainer board
2. DC power supply
3. Digital multimeter
4. DC Voltmeter
5. CD4007C IC (1 pc)
6. Resistor ($1K\Omega$ 1 pc)
7. Breadboard
8. Connecting wires

Circuit Diagram

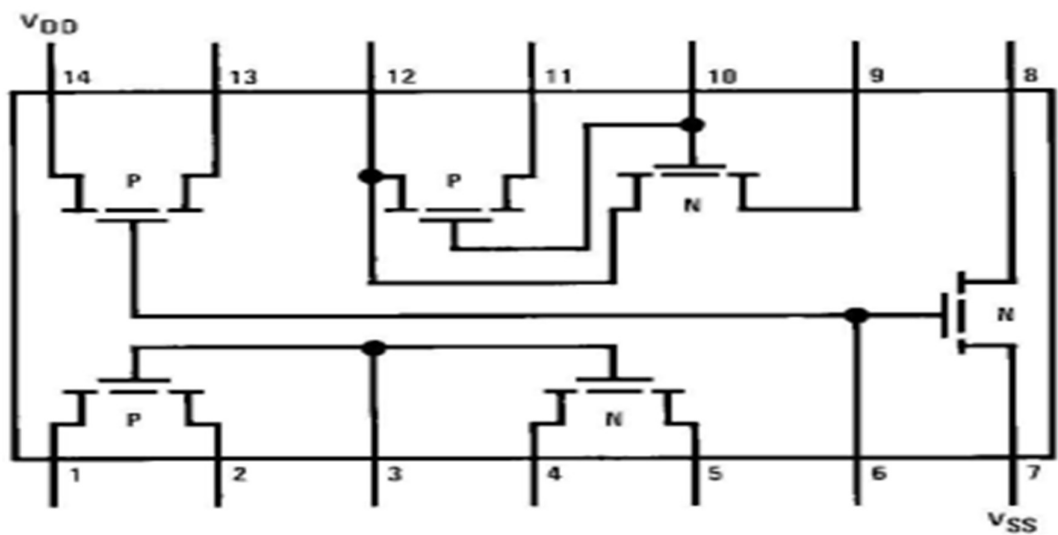


Figure-1: Pin diagram of CD4007C IC.

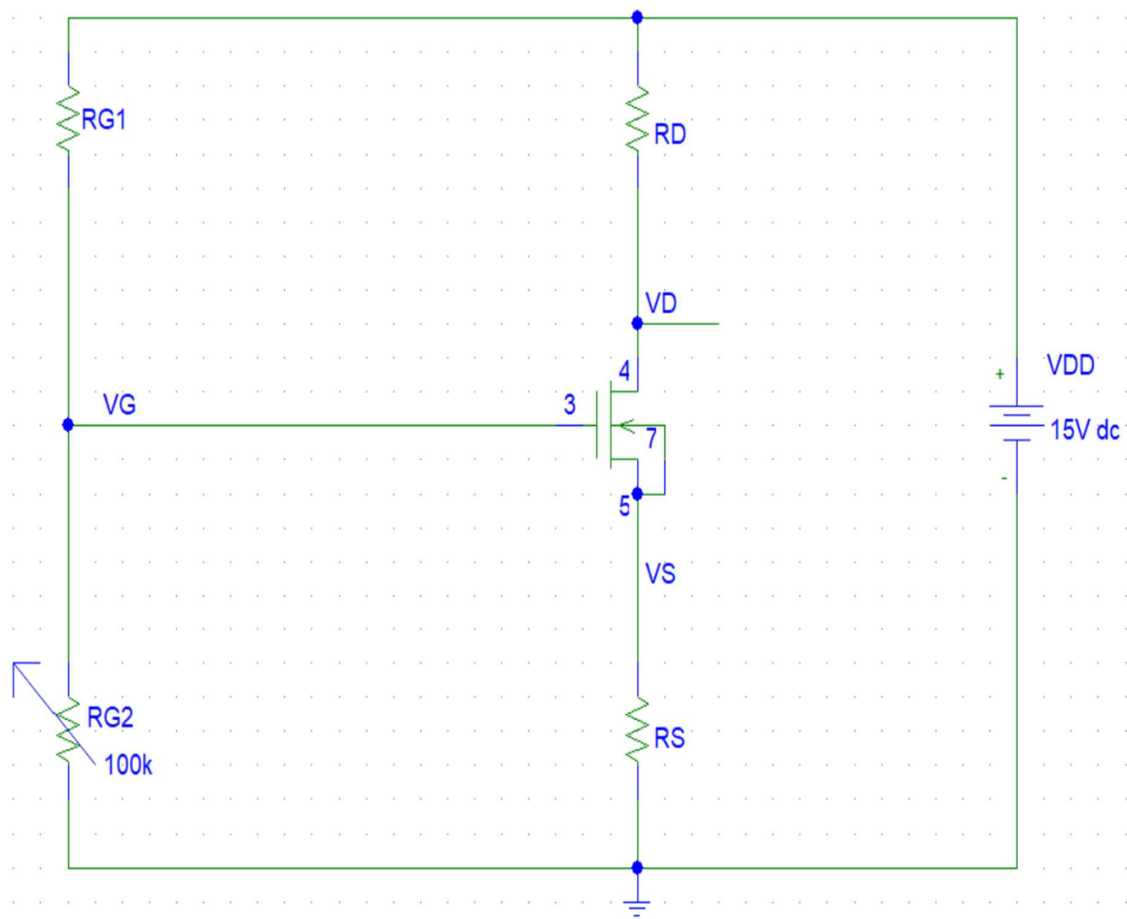


Figure-2

Pre-Lab

Here,

$$\text{Bias Current, } I_D = 0.6\text{mA}$$

$$\text{Source Voltage, } V_S = 4\text{V}$$

$$V_t = 1.2\text{V}$$

$$K_n \left(\frac{W}{L} \right) = 0.7\text{mA/V}^2$$

V_D is middle of V_S and V_{DD}

$$\therefore V_D = \frac{V_S + V_{DD}}{2} = \frac{4 + 15}{2} = 9.5\text{V}$$

$$\therefore R_D = \frac{V_{DD} - V_D}{I_D} = \frac{15 - 9.5}{0.6} = 9.167\text{K}\Omega$$

$$\therefore R_S = \frac{V_S}{I_D} = \frac{4}{0.6} = 6.667\text{K}\Omega$$

We know,

$$I_D = \frac{1}{2} K_n \left(\frac{W}{L} \right) (V_{GS} - V_t)^2$$

$$\therefore V_{GS} = \sqrt{\frac{2 \times I_D}{K_n \left(\frac{W}{L} \right)}} + V_t = \sqrt{\frac{2 \times 0.6}{0.7}} + 1.2 = 2.51\text{V}$$

$$\therefore V_G = V_{GS} + V_S = 2.51 + 4 = 6.51\text{V}$$

Using VDR,

$$V_G = \frac{R_{G_2}}{R_{G_1} + R_{G_2}} \times V_{DD}$$

$$\therefore R_{G_1} = \left(\frac{V_{DD}}{V_G} - 1 \right) R_{G_2} = \left(\frac{15}{6.51} - 1 \right) 100 = 130.4\text{k}\Omega$$

Total input resistance, $R_G = R_{G_1} \parallel R_{G_2}$

$$\therefore R_G = \frac{R_{G_1} \times R_{G_2}}{R_{G_1} + R_{G_2}} = \frac{130.4 \times 100}{130.4 + 100} = 56.6\text{K}\Omega$$

$$I_G = \frac{V_{DD}}{R_{G_1} + R_{G_2}} = \frac{15}{130.4 + 100} = 0.0651\text{mA}$$

$$\therefore I_{\text{total}} = I_D + I_G = 0.6 + 0.0651 = 0.6651\text{mA}$$

Total DC power dissipation,

$$P_{DC} = V_{DD} \times I_{\text{total}} = 15 \times 0.6651 = 99765\text{mW}$$

Experimental Datasheet

Sample No.	R_{G1} K Ω	R_D K Ω	R_S K Ω	R_{G2} K Ω	V_D (V)	V_S (V)	V_G (V)	V_{DS} (V)	I_D mA
1	97	8.621	6.165	70.145	4.82	3.5	5.127	1.32	0.5677
2				80.55	5.059	3.736	4.875	1.322	0.606
3				90.002	5.1	3.78	4.82	1.32	0.613

For sample 1,

$$I_D = \frac{V_S}{R_S} = \frac{3.5}{6.165} = 0.5667mA$$

For sample 2,

$$I_D = \frac{V_S}{R_S} = \frac{4.875}{6.165} = 0.606mA$$

For sample 3,

$$I_D = \frac{V_S}{R_S} = \frac{4.82}{6.165} = 0.613mA$$

Discussion

The objective was to understand and practically implement the biasing of an ~~NH~~ NMOS transistor for a common-source (CS) voltage amplifier, and design a biasing network that places a MOSFET in the saturation region, where it operates as a linear voltage amplifier.

The pre-lab calculation values for $V_D = 9.5V$, $V_{DD} = 15V$, $V_S = 4V$ and calculated values in the pre-lab for R_D and R_S were approximately $9.167k\Omega$ and $6.667k\Omega$ respectively.

Upon conducting three samples, the measured values were very close to our theoretical pre-lab values.

The minor differences between the calculated and experimental values can be attributed due to the actual values of the resistors used may not be exact, the CD4007C IC itself has variations in its transistor parameters ($K_n(\frac{W}{L})$ and V_{th}).

Despite the minor discrepancies, the experiment successfully demonstrated the principles of biasing a common source voltage amplifier. A key takeaway of this experiment is, if the gate voltage is too low, the MOSFET remains in the cutoff region and no amplification occurs. If too high, it may enter the triode ~~reg~~ region or draw excessive current, leading to distortion or even device failure. Further study for designing more complex analog circuits like multi-stage amplifiers and signal processing system.